Web Technologies

Lab Midterm

Submitted By:

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Submitted To:

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Table

N	BP	H	K	Sticky offset	Premium limit
7	607px	49	4	77px	Rs 700

Screenshots

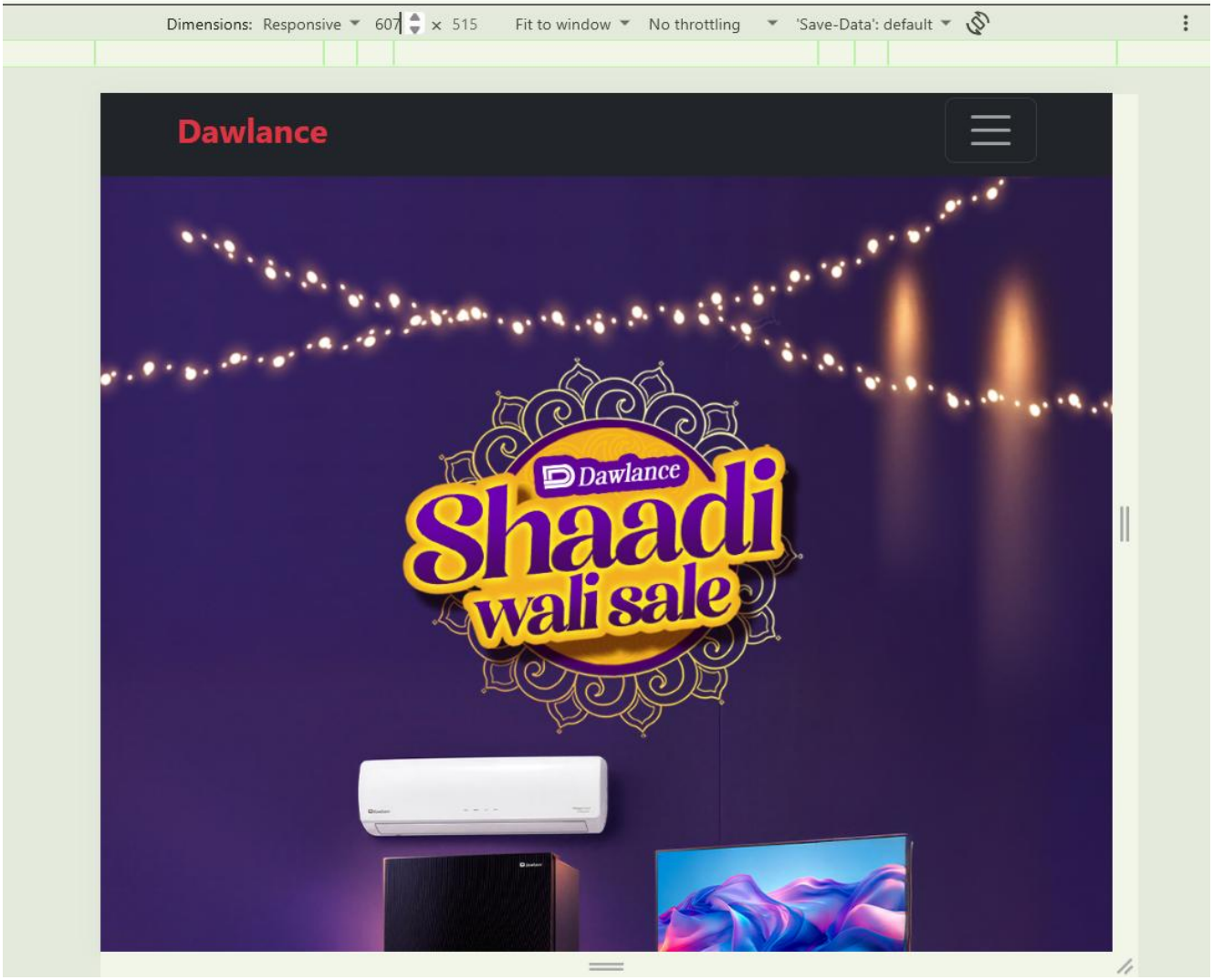
Replace each box with your screenshot. The text in the box says which page to open, which width to use and what must be visible.

Task 1 - CSS-only category filter (page: index.html)

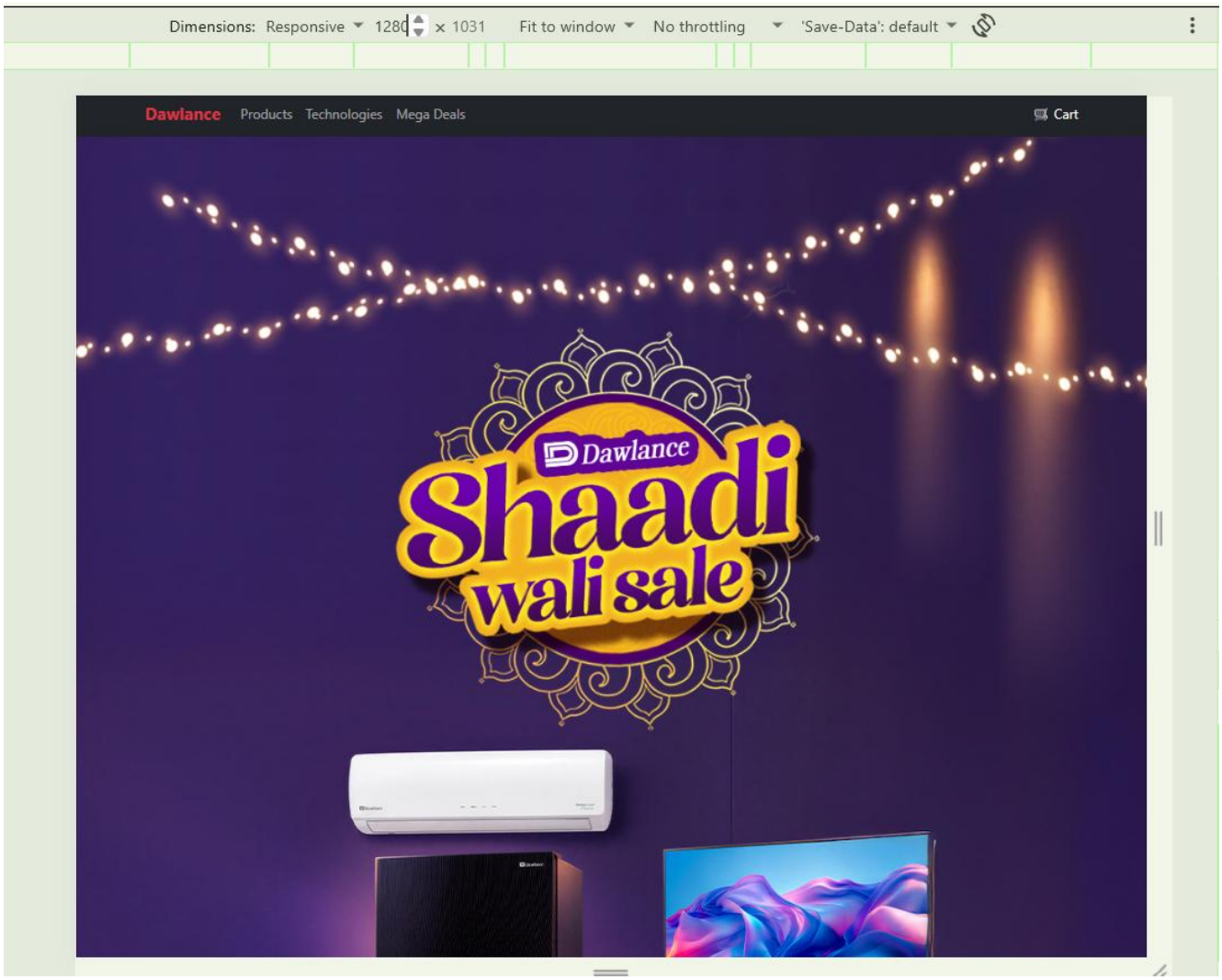
375 px



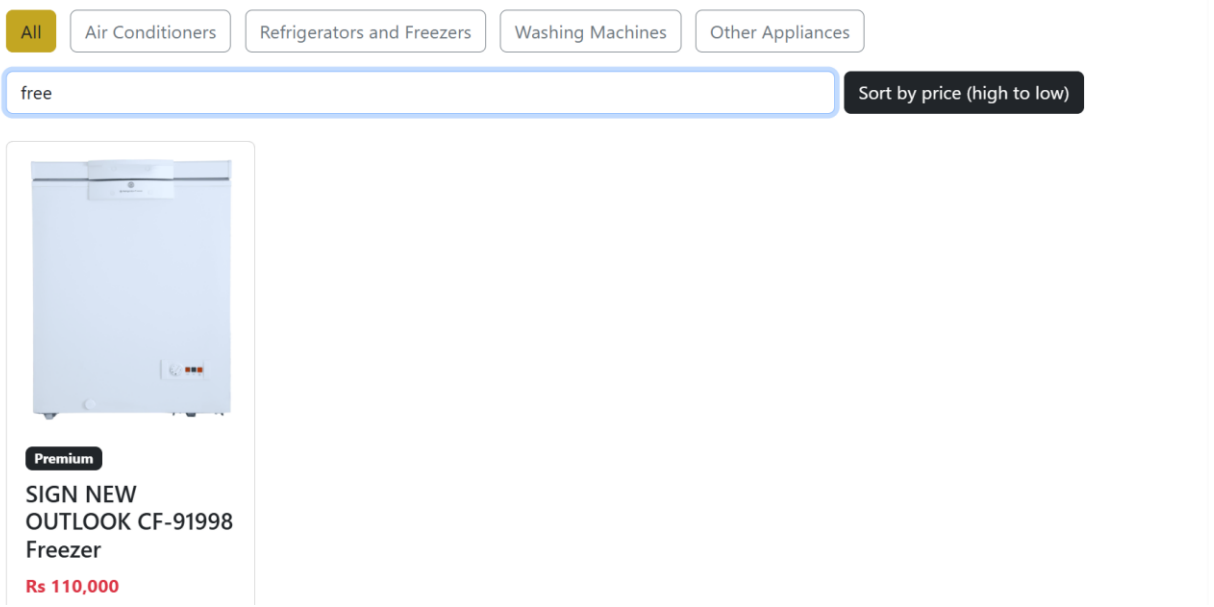
607 px


Dimensions: Responsive 607 x 515 Fit to window No throttling 'Save-Data': default
Dawlance
Dawlance
Shaadi
wali sale

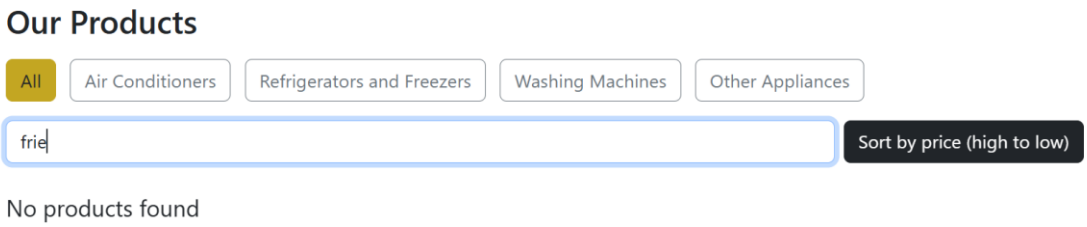
1280 px



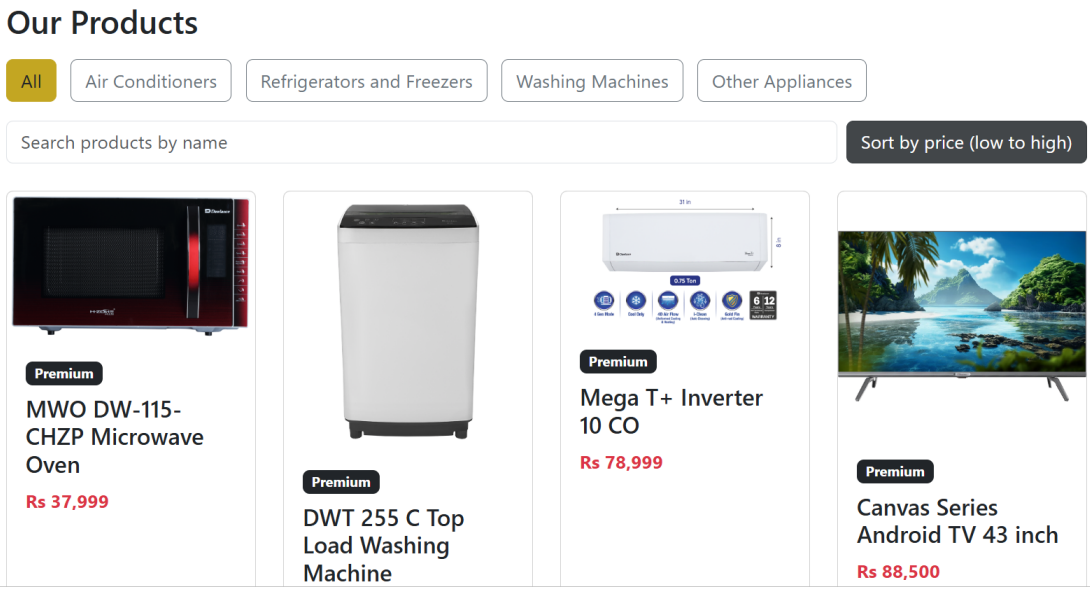
Searching Filter



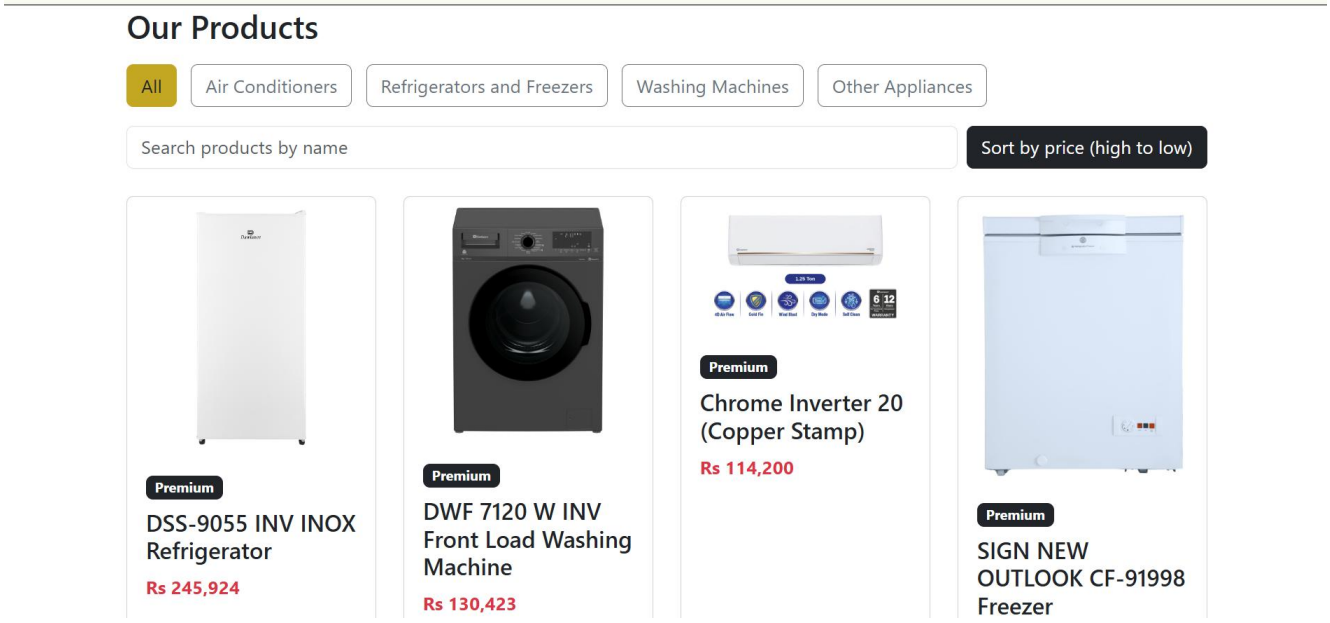
No Results Found



Low to High sorted



High to low Sorted



Code

Task 1 - CSS-only category filter

index.html (added above the grid; each card column also got data-category="ac", "fridge", "washer" or "other", and the grid got the class product-grid)

```
      <!-- ===== TASK 1: CSS-only category filter =====            The radio buttons MUST come
before the grid (same parent), because the CSS rule           uses the ~ selector to reach the
cards that come after the checked radio button. -->       <input type="radio" class="btn-check"
name="category" id="cat-all" value="all" checked>
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-all">All</label>
      <input type="radio" class="btn-check" name="category" id="cat-ac" value="ac">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-ac">Air Conditioners</label>
      <input type="radio" class="btn-check" name="category" id="cat-fridge" value="fridge">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-fridge">Refrigerators and
      Freezers</label>
      <input type="radio" class="btn-check" name="category" id="cat-washer" value="washer">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-washer">Washing Machines</label>
      <input type="radio" class="btn-check" name="category" id="cat-other" value="other">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-other">Other Appliances</label>
```

Style.css

```
/* ======================================================================
   MY NUMBERS (roll no FA24-BCS-007): N = 7, BP = 607px, H = 49, K = 4
   ====================================================================== */

/* Accent colour = hsl(H, 70%, 45%) with H = 49 */
:root {
  --accent: hsl(49, 70%, 45%);
}

/* ===== TASK 1: CSS-only category filter ===== */

/* The selected filter button gets the accent colour (the radio button is hidden by Bootstrap's btn-check) */
.btn-check:checked + .filter-label {   background-color: var(--accent);   border-color: var(--accent);
color: #212529; }

/* When a category radio button is :checked, hide every card (column) of the grid that is NOT in that category.
display: none removes the card completely, so the grid has no empty gaps. */ #cat-ac:checked ~ .product-grid
.col:not([data-category="ac"]),
#cat-fridge:checked ~ .product-grid .col:not([data-category="fridge"]),
#cat-washer:checked ~ .product-grid .col:not([data-category="washer"]),
#cat-other:checked ~ .product-grid .col:not([data-category="other"]) {
display: none; }
```

Task 2 - Product page layout with a sticky details box

product.html (main part)

```
<main class="container my-4">
```

```
    <!-- TASK 2: CSS Grid with named areas (see .product-layout in Style.css) -->
    <div class="product-layout">

      <!-- Area "image": the larger product image -->
      <div class="product-image">         <img
src="https://www.dawlance.com.pk/media/resize/F01000049100_MDM2_LOW_1.png/1000Wx1000H/image.webp"
class="img-fluid" alt="Mega T+ Inverter 10 CO">       </div>

      <!-- Area "details": name, price, quantity and Add to Cart (this box is sticky) -->
      <div class="product-details">
        <h1>Mega T+ Inverter 10 CO</h1>
        <p class="text-danger fw-bold fs-3">Rs 78,999</p>

        <!-- Quantity selector -->
        <label for="quantity" class="form-label">Quantity:</label>         <input type="number"
id="quantity" value="1" min="1" max="10" class="form-control mb-3">

        <!-- Add to Cart button (does not work yet) -->
<button class="btn btn-danger btn-lg">Add to Cart</button>
      </div>
      <!-- Area "description": long text, so the page scrolls and the sticky box can be seen working -->
<section class="product-description">
        <h2 class="h4">Description</h2>         <p>The Mega T+ Inverter 10 CO is an inverter split air conditioner
made for hot summers. Its inverter       compressor changes speed instead of switching on and off again and again, so
the room stays at a steady       temperature and the electricity bill stays lower.</p>         <p>4D Airflow moves
the air left and right and also up and down, so the cool air reaches every corner of the       room instead of one
spot. This is useful in larger bedrooms and lounges where one fixed direction leaves some       places warm.</p>
<p>The Memory Resumption feature remembers the last settings of the AC. After a power cut, the AC starts       again
with the same mode and temperature, so you do not have to set it again with the remote every time the       light
comes back.</p>         <p>Gen Mode lets you choose the current level the AC is allowed to draw. When you run the
house on a       generator or a small UPS, this helps you control the power use so the AC does not overload it.</p>
<p>The 6-in-1 health filters help to keep the air in the room cleaner. Dust and small particles are caught by
the filters, which is good for families with children or people who are sensitive to dust.</p>         <p>Zone Follow
uses a sensor in the remote control. The AC then cools by the real temperature around the       person holding the
remote and not only by the temperature near the indoor unit.</p>         <p>Auto cleaning of the AC interior removes
dust from the inside of the indoor unit by itself. This keeps the       cooling strong for longer and reduces the
number of times the unit needs to be cleaned by hand.</p>         <p>The AC comes with a 12 years compressor
warranty, 6 years warranty on functional parts and free       installation. Installments are available, and you can
get an additional 2.5% off if you pay by card.</p>       </section>

    </div>
  </main>
```

Style.css

```
/* ===== TASK 2: product page layout with CSS Grid (named areas) ===== */

/* Below 607px: one column, stacked: image, then details, then description */
.product-layout {   display: grid;   grid-template-columns: 1fr;   gap: 24px;
grid-template-areas:     "image"
    "details"
    "description";
}

.product-image {
grid-area: image; }

.product-details {   grid-
area: details;   border:
1px solid #ddd;   border-
radius: 10px;   padding:
20px;   background-color:
white; }

.product-description {
grid-area: description; }

/* From 607px (my BP) and wider: image and details side by side, description full width underneath */
@media (min-width: 607px) {
.product-layout {     grid-
template-columns: 3fr 2fr;
grid-template-areas:       "image
details"       "description
description";
  }

  /* The details box stays visible while scrolling: 77px from the top (N mod 40 + 70 = 7 + 70).
     align-self: start stops the grid from stretching the box, so it is free to stick. */
.product-details {     position: sticky;     top: 77px;     align-self: start;   }
}
```

Task 3 - JavaScript: live search and price sort

index.html (search box, sort button and message added above the grid; each price got the class price; script linked before </body>)

```
      <!-- ===== TASK 3: search box and sort button ===== -->
      <div class="d-flex gap-2 mb-4">
        <input type="text" id="search-box" class="form-control" placeholder="Search products by name">
<button type="button" id="sort-button" class="btn btn-dark text-nowrap">Sort by price</button>
      </div>

      <!-- TASK 3: shown instead of the grid when no card matches -->
      <p id="no-products" class="d-none fs-5">No products found</p>


<script src="script.js"></script>
```

script.js (new file)

```
// ===== TASK 3: live search, price sort and Premium labels =====
// My roll number is FA24-BCS-007, so N = 7. // Products costing
more than N x 100 = 700 get a "Premium" label. const N = 7; const
premiumLimit = N * 100;

// Find the elements on the page that the script needs const grid = document.querySelector(".product-
grid");              // the grid that holds the cards const cards = grid.querySelectorAll(".col");
// every product card (column) const searchBox = document.getElementById("search-box");           // the
search text box const sortButton = document.getElementById("sort-button");         // the "Sort by price"
button const noProductsMessage = document.getElementById("no-products");  // the "No products found"
message const categoryRadios = document.querySelectorAll('input[name="category"]'); // the 5 filter radio
buttons

// Read the price from the card itself, for example "Rs 78,999" becomes the number 78999
function getPrice(card) {   const priceText = card.querySelector(".price").textContent;
return Number(priceText.replace(/[^0-9]/g, ""));   // remove everything that is not a digit }

// Read the product name from the card, in lower case function
getName(card) {   return card.querySelector(".card-
title").textContent.toLowerCase(); }

// Show only the cards that match BOTH the search text and the selected category
function showMatchingCards() {   const searchText =
searchBox.value.toLowerCase().trim();   const category =
document.querySelector('input[name="category"]:checked').value;   let visibleCount
= 0;

  cards.forEach(function (card) {     const nameMatches =
getName(card).includes(searchText);     const categoryMatches = category === "all" ||
card.dataset.category === category;

    if (nameMatches && categoryMatches) {       card.style.display = "";        //
empty value: the CSS decides (card is shown)       visibleCount++;     } else {
card.style.display = "none";    // hide the card     }
  });

  // If no card matches, hide the grid and show the message instead
if (visibleCount === 0) {     grid.classList.add("d-none");
noProductsMessage.classList.remove("d-none");   } else {
grid.classList.remove("d-none");
noProductsMessage.classList.add("d-none");   }
}

// Run the filter every time the user types, and every time a category is chosen
searchBox.addEventListener("input", showMatchingCards);
categoryRadios.forEach(function (radio) {   radio.addEventListener("change",
showMatchingCards); });

// Sort by price: first click low to high, next click high to low, and so on
let lowToHigh = true;

sortButton.addEventListener("click", function () {   // Put
the existing cards in an array and sort it by price   const
sortedCards = Array.from(cards).sort(function (a, b) {     if
(lowToHigh) {       return getPrice(a) - getPrice(b);     }
else {
      return getPrice(b) - getPrice(a);
}
  });
  // appendChild moves each existing card to the end of the grid, so the grid ends up in the sorted order
sortedCards.forEach(function (card) {     grid.appendChild(card);   });
```

```
  // Show the current order on the button, then switch the direction for the next click
if (lowToHigh) {     sortButton.textContent = "Sort by price (low to high)";   } else {
sortButton.textContent = "Sort by price (high to low)";   }   lowToHigh = !lowToHigh;
});

// When the page loads: add a "Premium" label to every card that costs more than the limit
cards.forEach(function (card) {   if (getPrice(card) > premiumLimit) {     const label =
document.createElement("span");     label.className = "badge bg-dark mb-2 align-self-start";
label.textContent = "Premium";     card.querySelector(".card-body").prepend(label);   // put the
label at the top of the card text   }
});

// Make sure the cards match the search box and radio buttons when the page first loads
showMatchingCards();
```

Complete updated index.html

The whole homepage file after Task 1 and Task 3 (filter buttons, search box, sort button, data-category on the cards and the script link).

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <!-- Makes the page fit the phone screen width (needed for responsiveness) -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dawlance - Home</title>

  <!-- Bootstrap CSS (framework) -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet">
  <!-- My own CSS -->
  <link rel="stylesheet" href="Style.css">
</head>
<body>

  <!-- ========== HEADER (same on both pages) ========== -->
  <header>
    <!-- Bootstrap navbar: full bar from 768px, hamburger menu below 768px -->
    <nav class="navbar navbar-expand-md navbar-dark bg-dark">
      <div class="container">

        <!-- Site name / logo on the left -->
        <a class="navbar-brand fw-bold text-danger" href="index.html">Dawlance</a>

        <!-- Hamburger button (only shows on small screens) -->
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#menu">
          <span class="navbar-toggler-icon"></span>
        </button>

        <!-- These links collapse into the hamburger menu on small screens -->
        <div class="collapse navbar-collapse" id="menu">
          <ul class="navbar-nav me-auto">
            <li class="nav-item"><a class="nav-link" href="index.html">Products</a></li>
            <li class="nav-item"><a class="nav-link" href="#">Technologies</a></li>
<li class="nav-item"><a class="nav-link" href="#">Mega Deals</a></li>
          </ul>

          <!-- Cart icon (does not work yet) -->
          <a class="nav-link text-white" href="#">■ Cart</a>
        </div>

      </div>
    </nav>
  </header>

  <main>

    <!-- Banner image under the header -->
    <img src="https://www.dawlance.com.pk/medias/Dawlance-Wedding-Vati-KV-final.jpg.webp?context=bWFzdGVyfGltYWdlc3wxM
      Dg4NTg0fGltYWdlL3dlYnB8YUdVekwyZzBPUzh4TXpjeU1ETXlOemMwT1RZMk1pOUVZWGRzWVc1alpWOVhaV1JrYVc1bklGWmhkR2xmUzFaZlptb
HVZV3d1YW5CbkxuZGxZbkF8ZjFiYTdhY2RkZWNiMGE3Y2I5ODFmOTE2ZTk0ZmFkMDQ0OTJlYTY0NTVlOTFhNDhhZGZiODU3YWY3YTcxZTE1OQ"
class="img-fluid w-100" alt="Dawlance banner">

    <section class="container my-4">
      <h1 class="h3 mb-3">Our Products</h1>

      <!-- ===== TASK 1: CSS-only category filter =====            The radio buttons MUST come
before the grid (same parent), because the CSS rule            uses the ~ selector to reach the
cards that come after the checked radio button. -->       <input type="radio" class="btn-check"
name="category" id="cat-all" value="all" checked>
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-all">All</label>
      <input type="radio" class="btn-check" name="category" id="cat-ac" value="ac">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-ac">Air Conditioners</label>
      <input type="radio" class="btn-check" name="category" id="cat-fridge" value="fridge">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-fridge">Refrigerators and
      Freezers</label>
      <input type="radio" class="btn-check" name="category" id="cat-washer" value="washer">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-washer">Washing Machines</label>
      <input type="radio" class="btn-check" name="category" id="cat-other" value="other">
      <label class="btn btn-outline-secondary filter-label me-2 mb-3" for="cat-other">Other Appliances</label>


      <!-- ===== TASK 3: search box and sort button ===== -->
      <div class="d-flex gap-2 mb-4">
        <input type="text" id="search-box" class="form-control" placeholder="Search products by name">
```

```
        <button type="button" id="sort-button" class="btn btn-dark text-nowrap">Sort by price</button>
      </div>

      <!-- TASK 3: shown instead of the grid when no card matches -->
      <p id="no-products" class="d-none fs-5">No products found</p>

      <!-- Bootstrap grid:            row-cols-1  = 1 card per row on phones
row-cols-md-2 = 2 cards per row from 768px (tablet)            row-cols-lg-4 =
4 cards per row from 992px (desktop)            g-4 = gap between cards -->
<div class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4 product-grid">

      <div class="col" data-category="ac">
        <div class="card h-100">
          <a href="product.html"><img src="https://www.dawlance.com.pk/media/resize/F01000049100_MDM2_LOW_1.png/1000Wx
      1000H/image.webp" class="card-img-top" alt="Mega T+ Inverter 10 CO"></a>
          <div class="card-body d-flex flex-column">
            <h5 class="card-title">Mega T+ Inverter 10 CO</h5>
            <p class="price text-danger fw-bold">Rs 78,999</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
<button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="ac">
        <div class="card h-100">
          <a href="product.html"><img src="https://www.dawlance.com.pk/media/resize/F01000028200_MDM2_LOW_1.png/1000Wx
      1000H/image.webp" class="card-img-top" alt="Chrome Inverter 20 (Copper Stamp)"></a>
          <div class="card-body d-flex flex-column">
            <h5 class="card-title">Chrome Inverter 20 (Copper Stamp)</h5>
            <p class="price text-danger fw-bold">Rs 114,200</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
<button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="fridge">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/freezer?lock=11" class="card-img-top"       alt="SIGN NEW OUTLOOK CF-91998
Freezer"></a>           <div class="card-body d-flex flex-column">
            <h5 class="card-title">SIGN NEW OUTLOOK CF-91998 Freezer</h5>
            <p class="price text-danger fw-bold">Rs 110,000</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
<button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="fridge">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/refrigerator?lock=12" class="card-img-top"       alt="DSS-9055 INV INOX
Refrigerator"></a>           <div class="card-body d-flex flex-column">
            <h5 class="card-title">DSS-9055 INV INOX Refrigerator</h5>
            <p class="price text-danger fw-bold">Rs 245,924</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
<button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="washer">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/washing,machine?lock=13"       class="card-img-top" alt="DWT
255 C Top Load Washing Machine"></a>           <div class="card-body d-flex flex-column">
<h5 class="card-title">DWT 255 C Top Load Washing Machine</h5>
            <p class="price text-danger fw-bold">Rs 50,552</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
<button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="other">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/microwave?lock=15" class="card-img-top"       alt="MWO DW-115-CHZP Microwave
Oven"></a>
          <div class="card-body d-flex flex-column">
            <h5 class="card-title">MWO DW-115-CHZP Microwave Oven</h5>
            <p class="price text-danger fw-bold">Rs 37,999</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
```

```
            <button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="other">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/television?lock=18" class="card-img-top"       alt="Canvas Series Android TV
43 inch"></a>           <div class="card-body d-flex flex-column">
            <h5 class="card-title">Canvas Series Android TV 43 inch</h5>
            <p class="price text-danger fw-bold">Rs 88,500</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
            <button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      <div class="col" data-category="washer">
        <div class="card h-100">           <a href="product.html"><img
src="https://loremflickr.com/400/400/washing,machine?lock=20"       class="card-img-top" alt="DWF
7120 W INV Front Load Washing Machine"></a>           <div class="card-body d-flex flex-column">
            <h5 class="card-title">DWF 7120 W INV Front Load Washing Machine</h5>
            <p class="price text-danger fw-bold">Rs 130,423</p>
            <!-- mt-auto pushes the button to the bottom of the card -->
            <button class="btn btn-danger mt-auto">Add to Cart</button>
          </div>
        </div>
      </div>

      </div>
    </section>

  </main>

  <!-- ========== FOOTER (same on both pages) ========== -->
  <footer class="site-footer">
    <div class="container">

      <!-- Footer columns (flexbox, see Style.css) -->
      <div class="footer-columns">
        <div>
          <h5>Products</h5>
          <ul>
            <li><a href="#">Air Conditioners</a></li>
            <li><a href="#">Refrigerators</a></li>
            <li><a href="#">Washing Machines</a></li>
            <li><a href="#">Microwave Ovens</a></li>
          </ul>
        </div>
        <div>
<h5>Corporate</h5>
          <ul>
            <li><a href="#">Life At Dawlance</a></li>
            <li><a href="#">Policies</a></li>
            <li><a href="#">News &amp;amp; Events</a></li>
          </ul>
        </div>
        <div>
          <h5>Customer Support</h5>
          <ul>
            <li><a href="#">FAQs</a></li>
            <li><a href="#">Return And Refund Policy</a></li>
            <li><a href="#">Contact Us</a></li>
          </ul>
        </div>
      </div>

      <p class="copyright">&copy; 2026 dawlance.com.pk</p>
    </div>
  </footer>

  <!-- Bootstrap JS: only needed so the hamburger button opens the menu -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
  <script src="script.js"></script>
</body>
</html>
```